P01

Stigmergic Coordination in Ant Colony Foraging

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Domain DOM_BIO

Abstract

We analyze how social insects coordinate complex foraging behaviors without centralized control. Through stigmergy, individual workers deposit pheromone cues in the environment, creating positive feedback loops that guide nestmates to high-reward food sources while evaporation allows dynamic rerouting.

Abstract quoted from G. Theraulaz et al., Artificial Life (1999).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?